Bridging the Mentorship Gap: Empowering Graduate Students to Guide Undergraduate Research Assistants

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1-1:50pm, Grand Mesa D

Rocky Mountain Psychological Association Annual Conference



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Why is it important for graduate students to mentor undergraduate research assistants (URA)?

- Many research studies in psychology need more hands.
- In large university settings, faculty members often lack the capacity to individually mentor all URAs.
- Benefits of developing mentorship skills for Graduate Students (GS) (Deng et al., 2024)
  - Career and professional development: project management, supervision, communication… Transferrable in both academic and industry careers
  - Leadership and confidence
    - You learn better when you teach
  - Personal growth and networking
    - Constructive feedback, patience and empathy, interpersonal dynamics, conflict resolutions…

Challenges

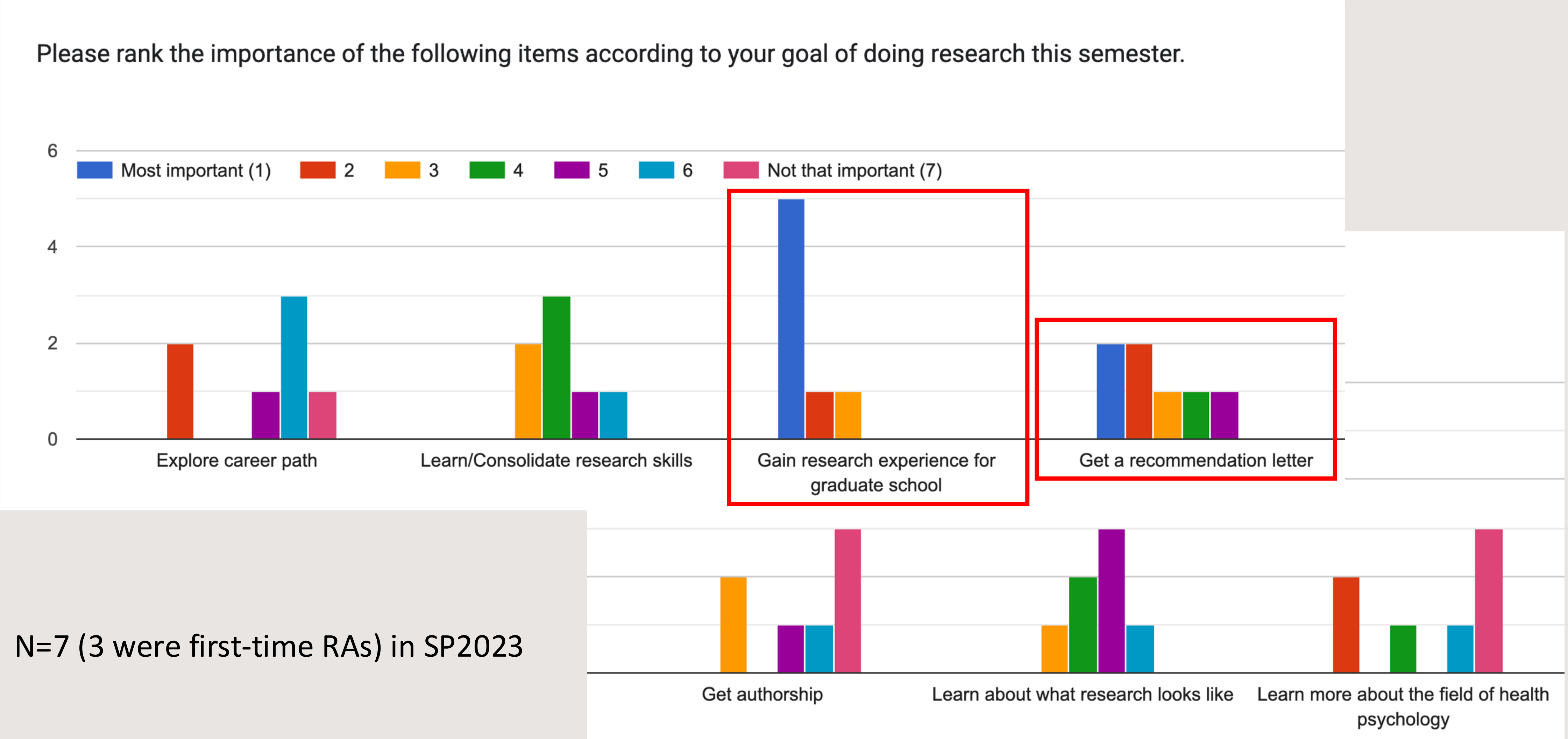
- Mentorship is not an autonomically acquired skill Lack of training and models (for both faculty and grad students) (Hund et al., 2018)
- Balancing multiple roles
- Time and workload
- Imposter syndrome and confidence

How to better mentor URAs?

#1: Goal Setting and Individualized Plans

- Why do URAs want to do research?
  - Interest? Grad school? Explore career paths? Personal influence?
- Clear purpose and research interest → content knowledge and specialized skills
- Just want to explore → transferrable skills (methodology, time management, communication, numeracy)
- Use Individual Development Plan (IDP) to help URAs to ***find*** their motivations
- People from different backgrounds might have different needs (Ishiyama, 2007; Mekolichick & Bellamy, 2012; Mekolichick & Gibbs, 2012)

What do URAs want?



N=7 (3 were first-time RAs) in SP2023

#2: Structured Mentor-Mentee Relationship

- URAs need a structure, like other undergraduate-level courses.
- Regular check-in meetings (group AND one-on-one)
- A syllabus (a roadmap)
  - Roles/responsibilities
  - Communication expectations
  - “Assignments”
  - Meeting content
- Help GS to stay on track as well

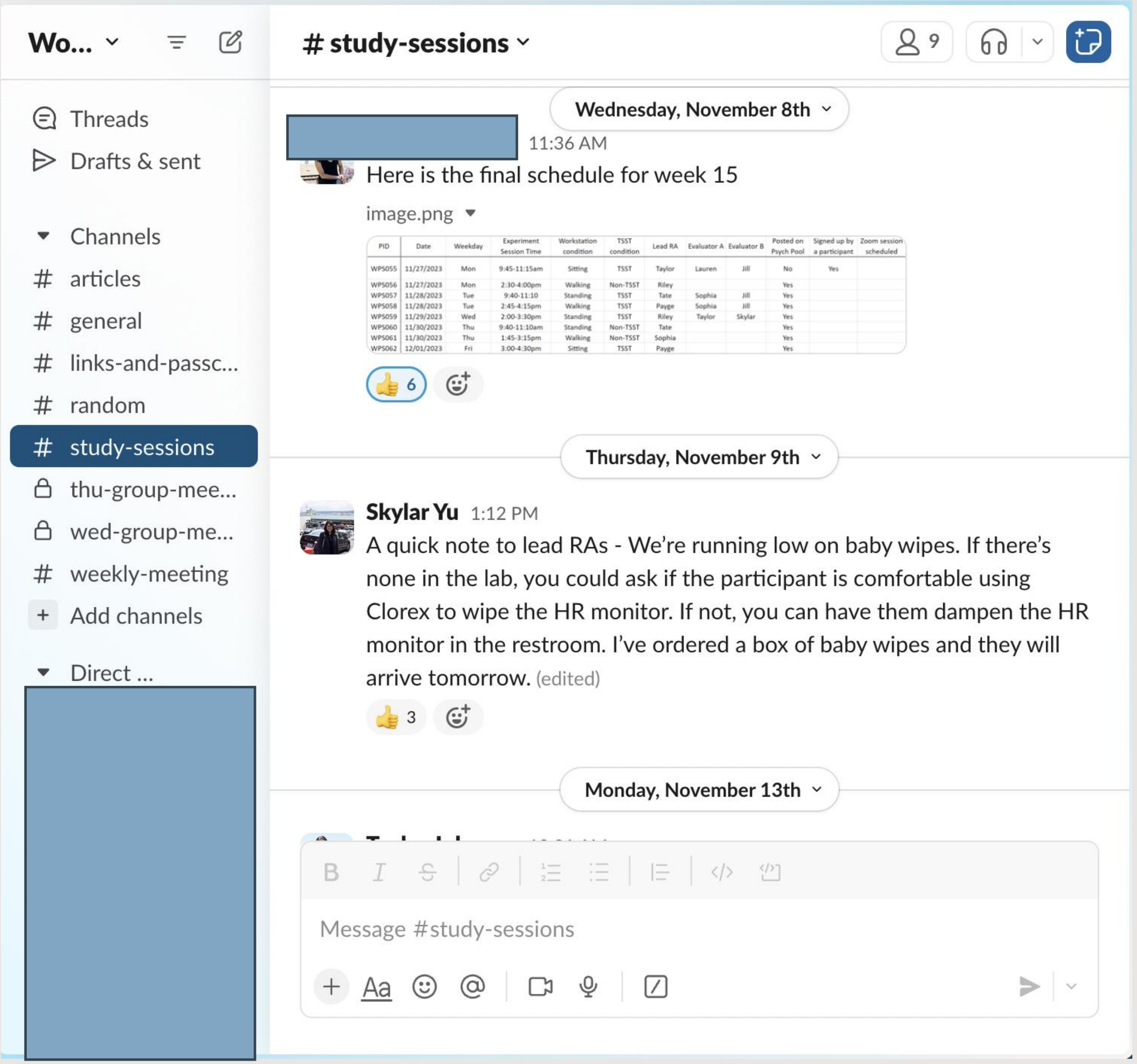
Syllabus

Week #	Date	Content	Discussion Leader / Presenter(s)
Week 1		**New RA's online training, returning RA completing assigned tasks**	
Week 2	8/30 8/31	**Introduction to group members and the study**	
Week 3	9/6 9/7	**Training**	
Week 4	9/13 9/14	**What's stress? How to measure it?** **Hellhammer, D. H., Stone, A. A., Hellhammer, J., & Broderick, J. (2010). Measuring stress. Encyclopedia of behavioral neuroscience, 2, 186-191.**	
Week 5	9/20 9/21	**Stress and Health I** **Slavich, G. (2022, April). Speaking of Psychology: How to keep stress from harming your health [Audio podcast episode]. Speaking of Psychology. Retrieved from https://www.apa.org/news/podcasts/speaking-of-psychology/stress**	
Week 6	9/27 9/28	**Stress and Health II** **Schneiderman, N., Ironson, G., & Siegel, S. D. (2005). Stress and health: psychological, behavioral, and biological determinants. Annu. Rev. Clin. Psychol., 1, 607-628.**	
Week 7	10/4 10/5	**Exercise and Stress** **Tsatsoulis, A., & Fountoulakis, S. (2006). The protective role of exercise on stress system dysregulation and comorbidities. Annals of the New York Academy of Sciences, 1083(1), 196-213.**	
Week 8	10/11 10/12	**RA Presentations I**	

Syllabus (cont.)

Week 9	10/18 10/19	**Light Physical Activity** **Amagasa, S., Machida, M., Fukushima, N., Kikuchi, H., Takamiya, T., Odagiri, Y., & Inoue, S. (2018). Is objectively measured light-intensity physical activity associated with health outcomes after adjustment for moderate-to-vigorous physical activity in adults? A systematic review. International Journal of Behavioral Nutrition and Physical Activity, 15(1), 1-13.**
Week 10	10/25 10/26	**Sedentary Behaviors** **Teychenne, M., Stephens, L. D., Costigan, S. A., Olstad, D. L., Stubbs, B., & Turner, A. I. (2019). The association between sedentary behaviour and indicators of stress: a systematic review. BMC public health, 19(1), 1-15.**
Week 11	11/1 11/2	**Active Workstations** **Dupont, F., Léger, P. M., Begon, M., Lecot, F., Sénécal, S., Labonté-Lemoyne, E., & Mathieu, M. E. (2019). Health and productivity at work: which active workstation for which benefits: a systematic review. Occupational and environmental medicine, 76(5), 281-294.**
Week 12	11/8 11/9	**RA Presentations II**
Week 13	11/15	**Explore Graduate Student Showcase** **https://wsnet2.colostate.edu/cwis478/GraduateStudentShowcase/**
Week 14		**Fall Recess**
Week 15	11/29 11/30	**RA Presentations III**
Week 16	12/6 12/7	**Q&A and Last Lab Fun!**

Communication



#3: Balancing Structure and Autonomy

- Start with clear guidelines on work ethics and procedure (Shanahan et al., 2015)
  - Reliability of URA is the key to help GS complete their research projects
- As URAs become more comfortable with the projects in the lab → more autonomy and leadership opportunities
  - Organizing shift arrangement, training new URAs, peer-mentoring, managing specific components, literature review, and more…
- Individual research project (can just be a plan)

#4: Providing a Supportive Environment

- Doing research can be intimidating for new URAs → timely, positive, and constructive feedback is crucial
- Acknowledge accomplishments and provide clear guidance on areas for improvement
- Provide an environment that supports honesty and sharing errors → a “blame-free” zone

Deviations (Lead RA)	Deviations (Evaluators)	Additional Notes
Before collecting T2 saliva sample the participant could not open the tube. Lead RA had to help participant open the tube	Evaluator A ended the first task instead of Evaluator B. During Free Speech task, Eval A asked a question not on the procedure packet	
first participant with the accelerometer		
None	Evaluator B ended the math task 25s early	
None		
participant asked how much time was left and if she was done during TSST task; HR reading temporarily stopped in the middle of T4	The participant cried during the TSST math task	
Lead RA accidently said "this will be our final round of measurements during T4", and then corrected themselves	zoom was muted in beginning. evaluator smirked when fixing	
participant tried making casual conversation immediately after restarting the video after each measurement; HR missed about the first minute of recording T5		participant kept fidgeting with BP cuff; participant came about 15 minutes early and asked if he had any papers to sign

Dr. Julia Strand

Strand, J. F. (2023). Error tight: Exercises for lab groups to prevent research mistakes. *Psychological Methods.* Advance online publication. https://doi.org/10.1037/met0000547

#5: Comprehensive Research Exposure

- Study development (literature reading and study design)
- Data collection and analysis
- Manuscript writing and authorship (this is tricky)
  - Publication process even for those who are not directly involved
- Local conferences and university-level symposium
- Rationales behind every decision
- A holistic understanding of research and develop ownership

Time Management for Grad Students

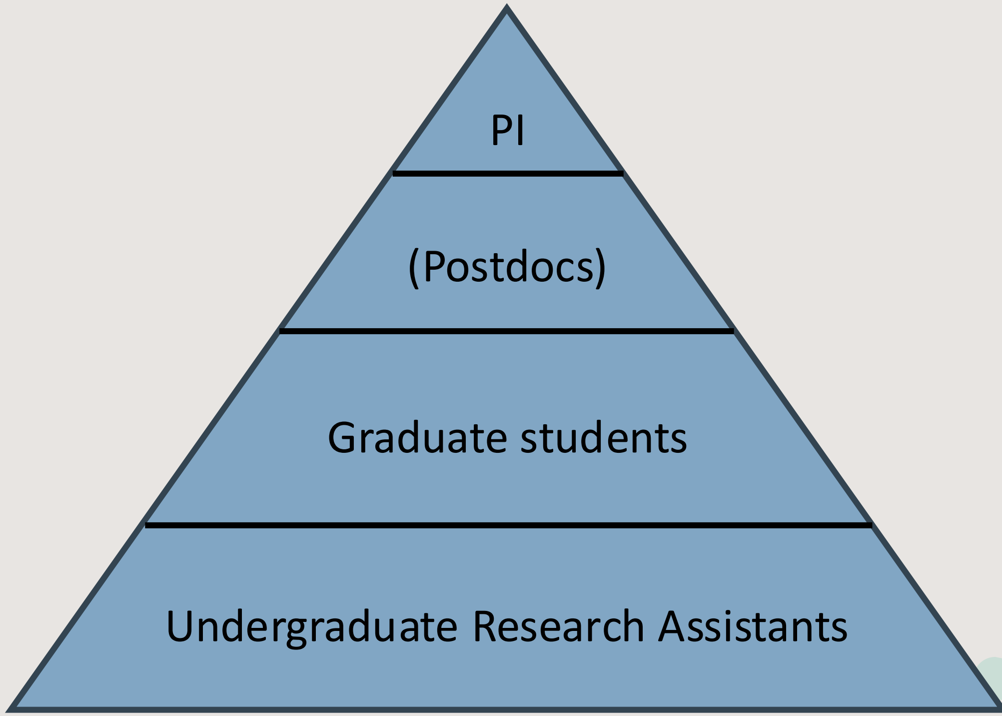
- Being a student + Being a researcher (+ Being a TA/Instructor)
- + Being a mentor
- Think about your priorities and the skills you want to develop
- Investing time in mentorship saves time in daily repeated tasks (experiment sessions, lab logistics, etc.) → Everyone benefits!
- Re-use the materials (e.g., turn the meeting content to a course later)

What can faculty do to empower GS mentorship?

- Provide **proactive** training and set expectations (like what people do for teaching)
  - Professional boundaries, appropriate conduct, mentorship strategies, conflict resolutions…
- Offer ongoing oversight and regular feedback
  - Joint mentoring session, debriefing after group meetings…
- Give space for personal growth and allow for mistakes
- Tiered mentoring
- Encourage GS to attend mentoring training
- Role model

Build Community Among Members of the Team

- PI meet at least once or twice per term with the team of URAs
  - Get to know the URAs
  - Rec letters
  - Different perspectives
- Fun group bonding activities
- Lab website



(Shanahan et al., 2015; Whiteside et al., 2007)

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Questions?

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